Macro Economics:

Classification of Economic Systems

on the basis of ownership of resources
- Capitalism
- Mixed System
- Socialism

On the basis of extent of market mechanism
- Market Economies
- Mixed Economies
- Planned Economies

Germany → East & West Germany
↳ war in 1989

Market Economy ⟹ Govt. on its own can't provide the best facilities
↳ Public – Private Partnership.

govt. can intervene with price ceilings, floors.

< 3% people pay income tax in India.

Adani Ports : Kerala, Vizag | Govt. will give the land, but can't build the ports

Petrol prices in India is no longer controlled by Govt.

PUBLIC INVESTMENTS:

Public Infrastructure: Roads, Railway Network, Highways, Airports

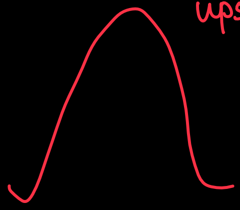
Agriculture

Industrial development ⟹ Jobs

1951-56 ⟹ agriculture (due to Bengal famine)
↳ first 5-year plan

Planning Commission (PC Mahaladun ..)
↳ Industry ⇒ IITs IIT Kharagpur

Washerman upstream → polluter ⇒ tax the polluter



Fisherman → polluted ⇒ subsidize the polluted.

Ronald Coase ⇒ Coase's Theorem → property rights.

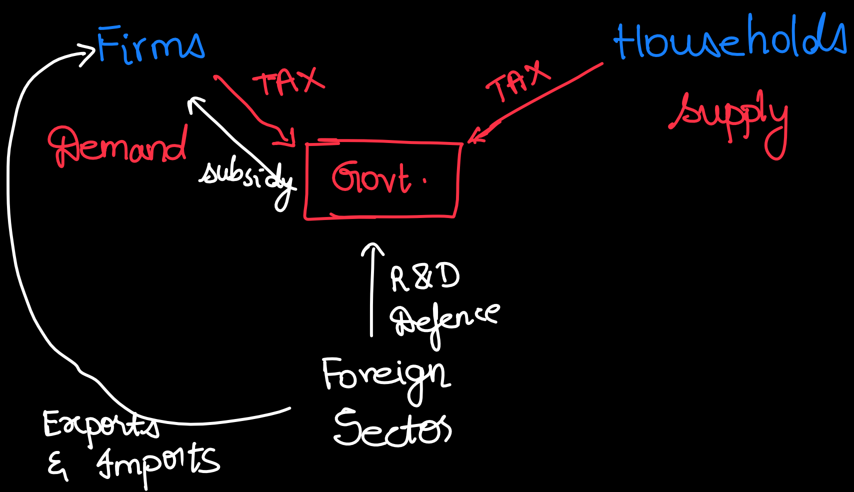
TR = price x Quantity

If P > MC, left up to the individual firm; how much mark-up (or) premium it will charge.

Vicious Circle of Poverty. ⇒

Low Income → Low Savings → Low Investment → Low Formation of Capital → Low Productivity → Low Income

Two - Sector Model



$$\text{Growth Rate} = \frac{GDP_t - GDP_{t-1}}{GDP_{t-1}} \times 100$$

National Income Accounting:

$$Y = C + I + G + (X - M)$$

Y ⇒ GDP (or) GNP

Y ⇒ output (Supply)

C ⇒ consumption demand ⇒ house-hold sector demand

I ⇒ private "gross" investment demand ⇒ business sector demand (Firms)

G ⇒ government expenditure/spending demand ⇒ government sector demand

X ⇒ Exports demand ⇒ Foreign demand for India's products

M ⇒ Imports demand ⇒ Home demand for foreign products

Exports demand, Imports demand ⇒ Foreign sector

C + I + G + (X − M) ⇒ # "Aggregate" Demand of Economy

$$GNP = GDP + NFIA$$

* Output is demand determined.

[$y = f(x)$: x is determining y.]

Export : earn Forex

Import : paying out

X − M ⇒ NX ⇒ net exports

{ C ⇒ largest component of aggregate demand }

C : anything that household sector buys / demands. ⇒ anywhere in the world

Depreciation ⇒ wear and tear of capital stock

a part of the capital stock gets wear and tear.

$$\boxed{\text{Net Investment} = \text{Gross Investment} - \text{Depreciation}}$$

Gross

Depreciation ⇒ sign can be +ve or −ve ⇒ real-estal

NDP ⇒ net domestic product

⇒ NDP = GDP − depreciation

GDP ⇒ gross domestic product

NNP ⇒ net national product

⇒ NNP = GNP − depreciation

GNP ⇒ gross national product.

*

House Hold Sector $\xrightarrow{\text{spends}}$ Business Sector

Business Sector $\xrightarrow{\text{wages/rents/salaries}}$ House Hold Sector

when CLOSED economy w/o Govt. $\Rightarrow$ $\boxed{Y = C + I}$

no exports ; no imports $\Rightarrow$ no interaction with the rest of the world

Summer of 1991 (June) $\Rightarrow$ India became open economy

↳ opened doors for exports & imports.

Pre-1991 $\Rightarrow$ $Y = C + I + G$

bandwagen effect

If in ~~monarchy~~ $\Rightarrow$ $Y = C + I$

no role of government in economy

Govt. $\Rightarrow$ only maintaining law and order

Market forces attain equilibrium

$$\frac{dP}{dt} = \lambda\left[D(P) - S(P)\right]; \ \lambda > 0$$

$$Y - C = I$$

$$\Rightarrow \boxed{S = I}$$

Saving - Investment Equality

equilibrium in MACRO - ECONOMICS

After introducing : role of govt. ; foreign sector

$$S - I = \underbrace{(G + T_R - T_X)}_{\text{domestic (internal)}} + (X - M) \Rightarrow \text{open economy.}$$

$$\Rightarrow \boxed{S - I \neq 0}$$

mismatch

$$\begin{bmatrix} S < I \\ S > I \end{bmatrix}$$

An open economy context : S-I comprises of 2 very different components

$$S - I = (G + T_R - T_x) + (X - M)$$

result

leakage | "Injection" | Domestic Affairs | Foreign Affairs

$$Y = \boxed{C} + I + G + (X - M)$$

max. no. of nobel prizes

$$Y_d = Y + T_R - T_x = C + S$$

GDP ⇒ C + (.)

output is demand determined

largest component of aggregate demand

Behavioural Foundations { Consumption is a psychological behaviour }

↳ don't change when we become richer

most stable ⇝ non-volatile and largest component of aggregate demand.

Sir J.M. Keynes (1936) ⇝ Oct 29, 1929 ⇒ BLACK TUESDAY

$$C = f(Y) \quad ; \quad f' > 0$$

$Y_d = Y + T_R - T_x$ ⇒ in closed-economy & no active role of govt.

⇒ no T_R, T_x

⇒ $Y = Y_d$

$$C = f(Y_d)$$

Linear Form: $C = a + bY_d$

constant ⇒ autonomous consumption

$a > 0$

induced consumption (induced by INCOME)

Policy Trade-off

If High Employment ⇒ high level of Income

desirable ↲ ↳ GOOD

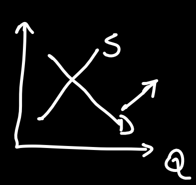
⇓

High Demand

⇓

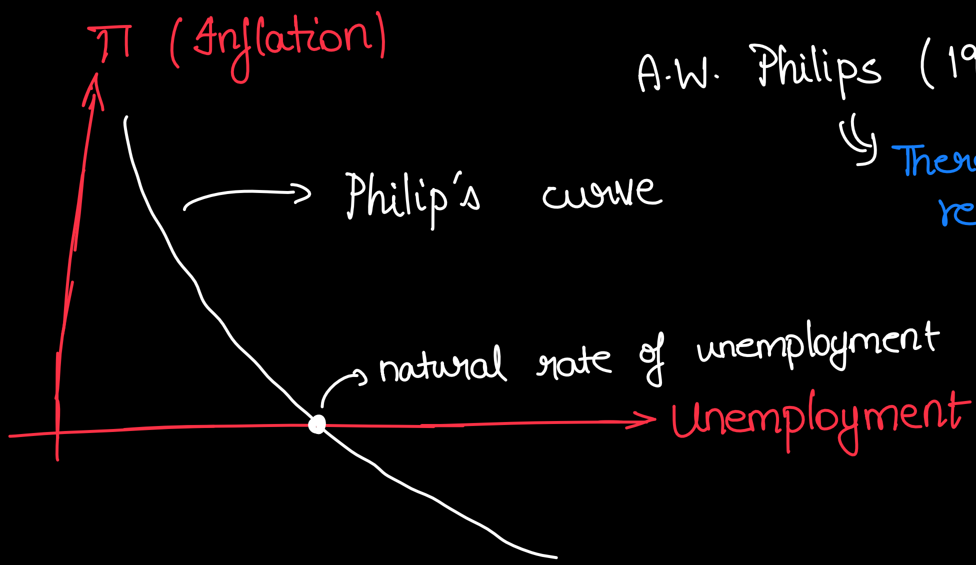
Inflation ⇐ increase in prices

undesirable ↲ ↳ BAD



* output always responds to DEMAND with a lag
↳ demand determined

#



A.W. Philips (1958)
↳ There is an INVERSE relationship.

High Employment level ⟶ Low Unemployment level

↳ High Inflation ⇄ INVERSE Relationship.

leakage injection

$$\underbrace{S - I}_{\text{saving investment gap}} = \underbrace{(G + T_R - T_x)}_{\text{Budget deficit surplus}} + \underbrace{(X - M)}_{\text{Trade Surplus Deficit}}$$

$$C = f(Y_d) = a + \boxed{bY_d}$$

C → consumption

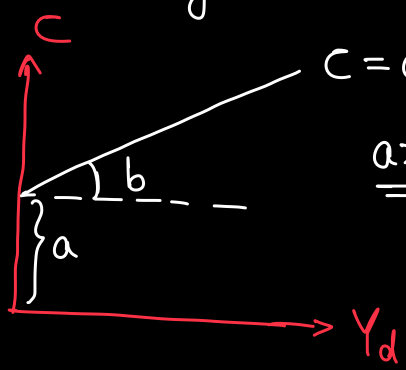
Y_d → current "real" disposable income

inflation adjusted ⇐

a → autonomous consumption

bY_d → induced consumption

$\boxed{0 < b < 1}$



$C = a + bY_d$; C↑ but less than proportionately

$\underline{\underline{a > 0}}$

$$\frac{\partial C}{\partial Y_d} = b$$

MPC ⟶ marginal propensity to consume

by The fundamental psychological law or'la Keynes (1936)

$0 < MPC < 1$ ⟶ proper fraction

{ high correlation coefficient b/w disp. income & household consumption }

$$\boxed{\frac{C}{Y_d} = \frac{a}{Y_d} + b}$$

b ⟶ MPC ⇒ $\boxed{APC - MPC > 0}$

delta increase in consumption per increase in unit income

APC : average propensity to consume ⟹ consumption per unit of disposable income.

* No role of interest rate in determining household consumption

* as Y_d↑ ⇒ APC↓

$$APC\downarrow = \frac{C}{Y_d} = \frac{\overset{\text{const.}}{a}}{Y_d\uparrow} + b$$

b ⟶ MPC

* Income explains Consumption (high correlation > 0.9

Simon Kuznets (1942